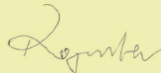




TEST REPORT IEC/EN 62109-1 Safety of Power Converter for use in Photovoltaic Power Systems Part 1: General requirements	
Report Number.:	GZES230100125901
Date of issue	2023-02-01
Total number of pages	72
Name of Testing Laboratory preparing the Report.....:	SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch
Applicant's name.....:	Zhejiang CHISAGE New Energy Technology Co., Ltd
Address	No. 1828 Fuqing South RD. Panhuo ST. Yinzhou District Ningbo Zhejiang 315000 China
Test specification: Standard.....: EN 62109-1:2010 IEC 62109-1:2010 (First Edition) Test procedure SGS-CSTC Non-standard test method.....: N/A	
Test Report Form No.....:	IEC62109_1B
Test Report Form(s) Originator.....:	VDE Testing and Certification Institute
Master TRF	Dated 2016-04
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Test item description	Single phase inverter
Trade Mark	 CHISAGE
Manufacturer	Zhejiang CHISAGE New Energy Technology Co., Ltd
Address	No. 1828 Fuqing South RD. Panhuo ST. Yinzhou District Ningbo Zhejiang 315000 China
Model/Type reference	See model list in page 7
Ratings	See model list in Page 7 to 8. Hardware version: Ver2.3 DC Software version: Ver0107 AC Software version: Ver2.5

Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒	Testing Laboratory:	SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch
Location/ address		198 Kezhu Road, Science City, Economic & Technology Development Area, Guangzhou, Guangdong, China
Tested by (name, function, signature)		Doris Tao (Project Engineer) 
Approved by (name, function, signature		Roger Hu (Technical Reviewer) 

List of Attachments (including a total number of pages in each attachment): 63 Page ~72 Page

1

Summary of testing:**Tests performed (name of test and test clause):**

The equipment has been tested according to the standard: EN 62109-1:2010 and IEC 62109-1:2010.

All test results are from the original report GZES210602035101, issued by SGS-CTS Standards Technical Services Co., Ltd Guangzhou Branch.

Testing location:

See page 2

Summary of compliance with National Differences (List of countries addressed):

No National Differences are addressed to this test report

Copy of marking plate(representative): 1

Utility-Interactive Inverter 2

Maximum input Voltage: 60Vdc 3

Range of input operating voltage: 25~55Vdc 4

Maximum input current: 13A x 4 5

Operating voltage range (AC): 230Vac 6

Rated output current(AC): 8.7A 7

Max output apparent power: 2000VA 8

Rated output power: 2000W 9

Rated AC Grid Frequency: 50/60Hz 10

Ambient Temperature: -40°C ~ +65°C 11

Peak efficiency: 96.5% 12

Protective class: Class I 13

Ingress protection: IP67 14

Max.Units per branch: 3 15

CHISAGE 17

Type: CE-1P20001G-230-EU 19

Number Serial: 20

CE 16

Warning symbols: 22, 23

Warnings: 24, 25, 26, 27, 28, 29, 30

www.chisagess.com 31

Note: 32

1. The above markings are the minimum requirements required by the safety standard. For the final production samples, the additional markings which do not give rise to misunderstanding may be added. 33
2. Label is attached on the side surface of enclosure and visible after installation
3. Labels of other models are as the same with **CE-1P20001G-230-EU**'s except the parameters of rating.
4. As declared by the applicant, the importer (and manufacturer, if it is different)'s name, registered trade name or registered trademark and the postal address will be marked on the products before being place on the market. The contact details shall be in a language easily understood by end-users and market surveillance authorities.

Test item particulars	Single Phase Inverter		
Equipment mobility	☐ movable ☒ fixed	☐ hand-held ☐ transportable	☐ stationary ☐ for building-in
Connection to the mains	☐ pluggable equipment ☒ permanent connection		☐ direct plug-in ☐ for building-in
Environmental category	☒ outdoor	☐ indoor unconditional	☐ indoor conditional
Over voltage category Mains	☐ OVC I	☐ OVC II	☒ OVC III ☐ OVC IV
Over voltage category PV	☐ OVC I	☒ OVC II	☐ OVC III ☐ OVC IV
Mains supply tolerance (%)	-90 / +110 %		
Tested for power systems	TN systems		
IT testing, phase-phase voltage (V)	N/A		
Class of equipment	☒ Class I ☐ Class II ☐ Class III ☐ Not classified		
Mass of equipment (kg)	3,5 kg for all model		
Pollution degree	Outside PD3; Inside PD2		
IP protection class	IP 67		
Possible test case verdicts:			
- test case does not apply to the test object.....: N/A			
- test object does meet the requirement.....: P (Pass)			
- test object was not evaluated for the requirement.....: N/E			
- test object does not meet the requirement.....: F (Fail)			
Testing			
Date of receipt of test item			
Date (s) of performance of tests: 2020-09-06 to 2020-09-28			

General remarks: 1

"(See Enclosure #)" refers to additional information appended to the report. 2
"(See appended table)" refers to a table appended to the report.

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and offenders may be prosecuted to the fullest extent of the law. Unless otherwise stated the results
shown in this test report refer only to the sample(s) tested.

Throughout this report a ☒ comma / ☐ point is used as the decimal separator. 4

Manufacturer's Declaration per sub-clause 4.2.5 of IEC 60335-1: 5

The application for obtaining a CB Test Certificate
includes more than one factory location and a
declaration from the Manufacturer stating that the
sample(s) submitted for evaluation is (are)
representative of the products from each factory has
been provided

6 ☐ Yes 7
☒ Not applicable

When differences exist; they shall be identified in the General product information section. 8

Name and address of factory (ies) 9 **NingBo Deye Inverter Technology Co., Ltd.** 10
No.26 South YongJiang Road, Daqi, Beilun, NingBo,
China.

General product information: 1

Product covered by this report is grid-connected PV inverter for indoor or outdoor installation. The connection to the DC input and AC output are through connectors. 2

The Solar inverter converts DC voltage into AC voltage. 3

The input and output are protected by varistors to Earth. The unit is providing EMC filtering at the output toward mains. The unit does not provide galvanic separation from input to output (transformerless). The output is switched off redundant by the high-power switching bridge and two relays. This assures that the opening of the output circuit can operate in case of single fault. 4

Equipment Under Testing: 5

CE-1P3001G-230-EU, CE-1P5001G-230-EU, CE-1P6001G-230-EU, CE-1P8001G-230-EU, CE-1P10001G-230-EU, CE-1P13001G-230-EU, CE-1P16001G-230-EU, CE-1P18001G-230-EU, CE-1P20001G-230-EU⁽¹⁾ 6

⁽¹⁾ Except for 4.2.2.6 TABLE: mains supply electrical data in normal condition is All models test, the other test clause model is CE-1P20001G-230-EU. 7

Model Number	CE-1P3001G-230-EU	CE-1P5001G-230-EU	CE-1P6001G-230-EU	CE-1P8001G-230-EU	CE-1P10001G-230-EU
Input (DC)					
Max. input power	400W	600W	800W	1200W	1200W
Max. input voltage	60V				
MPPT voltage range	25~55V				
Max. input current	13A	13A	13A×2	13A×2	13A×2
Output (AC)					
Rated grid voltage	230V				
Rated grid frequency	50Hz				
Rated output power	300W	500W	600W	800W	1000W
Rated output current	1.3A	2.2A	2.6A	3.8A	4.8A
Power factor	>0.99				
Ambient temperature	-40 °C ~ 65 °C				
Ingress protection	IP67				
Protective class	Class I				

Model Number	CE-1P13001G -230-EU	CE-1P16001G -230-EU	CE-1P18001G -230-EU	CE-1P20001G -230-EU	1
Input (DC)					
Max. input power	1600W	2400W	2400W	2400W	
Max. input voltage	60V				
MPPT voltage range	25~55V				
Max. input current	13A×4	13A×4	13A×4	13A×4	
Output (AC)					
Rated grid voltage	230V				
Rated grid frequency	50Hz				
Rated output power	1300W	1600W	1800W	2000W	
Rated output current	6.2A	7.7A	8.6A	9.6A	
Power factor	>0.99				
Ambient temperature	-40 °C ~65 °C				
Ingress protection	IP67				
Protective class	Class I				